Working Title: A Linear Immorality Direction in a Small Language Model

Author names here

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1 Introduction

Possible framing:

- Mechanistic interpretability aims to identify human-understandable variables inside language models.
- Prior work has found linear directions for sentiment and other properties, but morality-like representations are less settled, especially in small open models.
- The core question here is not whether Qwen behaves morally in some philosophical sense, but whether a simple, stable, causally relevant linear direction separates ETHICS scenarios labeled acceptable vs unacceptable/wrong.
- The paper's main claim can be modest: in Qwen2.5-0.5B-Instruct, a layer-15 residual direction fit from ETHICS short commonsense scenarios is stable, predictive, causally relevant, and only partly explained by sentiment.

Draft contribution list.

Something like:

- We identify a simple layerwise mean-difference direction in raw residual activations that separates ETHICS commonsense scenarios.
- We show that this direction is strongest around layer 15 and stable under both increasing sample size and bootstrap resampling.
- We show causal relevance by ablating the direction and measuring a substantial AUROC drop on moral judgment prompts.
- We compare this direction to a negative-sentiment direction learned from SST-2 and show moderate alignment but clear angular separation.
- We show that a sentiment-controlled residual morality direction still has causal force, and that random-direction ablations do not produce comparable effects.

2 Related Work

This section can stay short and tightly scoped. Likely subsections:

- Mechanistic interpretability and linear directions.
- Sentiment directions and representation engineering.
- Morality / ethics benchmarks in language models.
- Prior "moral direction" work and why the present paper is narrower and cleaner.

Use the literature note already in `notes/morality_direction_literature_note.md`.

2.1 Mechanistic Interpretability and Linear Directions

Mention work on sparse autoencoders, representation engineering, sentiment directions, and the broader idea that useful internal variables may be approximately linear in hidden-state space.

2.2 Moral and Ethical Evaluation in Language Models

Explain that ETHICS and related datasets are mostly behavioral benchmarks. The novelty here is not a new moral dataset, but using ETHICS as a source of high-consensus scenarios for hidden-state analysis.

2.3 Closest Prior Work

Be explicit and honest:

- broad claim "models have internal moral representations" is not new;
- the contribution here is a residual-stream, small-open-model, layerwise, causal, sentiment-controlled study using ETHICS commonsense short scenarios.

3 Experimental Setup

3.1 Model

Current model:

- `Qwen/Qwen2.5-0.5B-Instruct`
- 24 layers
- layer 15 selected by held-out ETHICS performance from a layer sweep

You can motivate the model as a small, cheap, open model suitable for rapid mechanistic experiments, while acknowledging its language/comprehension limits.

3.2 Moral Dataset

ETHICS commonsense short scenarios.

- Label 1: unacceptable / wrong / immoral in the dataset framing.
- Label 0: acceptable / not wrong.
- Training counts used in the main run: 2789 immoral, 3872 acceptable.
- Held-out eval: 983 immoral, 1126 acceptable.

Be careful with terminology. In prose, you can note that the dataset is about consensus moral acceptability judgments rather than deep moral philosophy.

3.3 Sentiment Control Dataset

SST-2, with the sign flipped so the learned control direction points toward negative sentiment. Matched-size train subset:

- 2789 negative
- 2789 positive

This gives a fairer comparison than letting sentiment use a much larger dataset.

3.4 Activation Readout

Use the final token of a fixed prompt template, e.g.

- ETHICS representation prompt: `Scenario: {text}\nMoral evaluation:`
- sentiment representation prompt: `Text: {text}\nSentiment:`

The rationale is that the final prompt token is a natural readout position where the preceding scenario has been integrated.

4 Methods

4.1 Direction Estimation

The simplest possible estimator:

$$d = \mu_{\text{immoral}} - \mu_{\text{acceptable}}$$

with d normalized to unit length. Stress the simplicity: no SAE, no nonlinear probe, no multi-token aggregation beyond the final pooled state.

4.2 Layerwise Sweep

Describe the layer sweep over raw residual activations. The main result:

- best ETHICS eval AUROC around layer 15
- nearby mid/late layers also strong

Possible figure: layerwise AUROC plot.

4.3 Held-Out Evaluation

The basic question is whether a direction fit on the train split separates acceptable vs immoral scenarios on held-out ETHICS examples. Likely metrics:

- AUROC as the primary metric
- accuracy as secondary, with a note that it is threshold-sensitive

4.4 Causal Intervention

Define the intervention clearly. If u is the learned unit direction and h the hidden state at the prompt boundary token, the removal intervention is:

$$h' = h - (h \cdot u)u$$

The judgment prompt scores `acceptable` vs `unacceptable`. Explain why AUROC is the right primary outcome: it measures separation, not threshold calibration.

4.5 Sentiment Control

Three comparisons belong here:

- full immorality direction
- pure negative-sentiment direction
- immorality residual after orthogonalizing against sentiment

Write down the orthogonalization:

$$u_\perp = \frac{u - (u \cdot s)s}{\|u - (u \cdot s)s\|}$$

where u is immorality and s is the sentiment direction.

4.6 Random-Direction Null

Important control: matched-norm random-direction ablations at the same layer and token position. This addresses the objection that "any disturbance" might hurt performance.

4.7 Bootstrap Direction Uncertainty

This section can be short but useful. Explain the stratified bootstrap over training activations:

- resample each class with replacement at original class size
- refit the mean-difference direction
- compare the bootstrap direction to the full-data direction via cosine and angle

This yields interpretable angular stability estimates for both immorality and negative sentiment, plus the uncertainty of their alignment.

5 Results

5.1 A Stable Linear Immorality Direction Exists

Core numbers to include:

- full ETHICS sweep best layer: 15
- eval AUROC around 0.829 for the direction-only classifier
- sample-size stabilization already visible as train size increases

Good figure candidates:

- layerwise AUROC
- sample-size stability curve

5.2 Removing the Direction Degrades Moral Separation

Use the full held-out causal numbers. Main AUROC result:

- baseline: 0.7758
- remove immorality direction: 0.6357

State plainly that this is evidence of causal relevance, not merely correlation.

5.3 Sentiment Explains Part, But Not All, of the Signal

Key numbers:

- remove pure sentiment: AUROC 0.6607
- remove sentiment-orthogonalized immorality residual: AUROC 0.7407
- cosine between full-data immorality and negative-sentiment directions: 0.5219
- angle between them: about 58.5 degrees

The wording here matters. They are substantially aligned relative to random high-dimensional directions, but robustly distinct.

5.4 Random Directions Do Not Reproduce the Effect

Main null result:

- morality removal AUROC drop: about 0.140
- 32 random-direction removals: mean drop near 0, max drop about 0.007

This is one of the strongest controls in the paper.

5.5 Bootstrap Stability

Key numbers:

- immorality direction stays within roughly 6 degrees of the full-data estimate under bootstrap resampling
- negative-sentiment direction also within roughly 6 degrees
- the alignment between them is itself stable

Use the note in `notes/bootstrap_uncertainty.md`.

5.6 Visualization and Error Analysis

Possible figure set:

- immorality vs residual PC1 scatter
- same plot with negative-sentiment comparison arrow

Possible analysis paragraph:

- highest-projection immoral examples are extreme violence / abuse / fraud
- high-projection acceptable examples often contain violent or taboo lexical content
- low-projection immoral examples are often milder, oddly phrased, or linguistically noisy

This suggests the direction captures real moral signal, but is strongly shaped by vivid harm-related language.

5.7 Latent-Output Mismatch

Include the result that the latent immorality projection aligns with ETHICS labels slightly better than Qwen's explicit one-word judgment. This is a nice payoff because it suggests that the latent direction can be cleaner than the model's expressed answer in a weak model.

6 Discussion

6.1 What Does the Direction Capture?

Good discussion point: the direction is neither a full moral theory nor just sentiment. It seems to combine:

- harm / violence
- taboo / crime semantics
- negative affect
- some additional moral signal beyond pure sentiment

6.2 Limitations

Likely limitations:

- small model with clear comprehension limitations
- one dataset and one prompt format
- linear direction is only an approximation
- ETHICS captures consensus labels, not universal morality
- residual-stream analysis alone does not identify a full circuit

6.3 Future Work

Natural next steps:

- repeat on Gemma
- extend to controversial items and latent/output comparisons
- connect the direction to SAE features or circuits
- compare pretrained vs instruct-tuned models

7 Conclusion

Keep the conclusion modest and crisp. Something like:

- there is a simple, stable, causally relevant linear direction in Qwen 0.5B that tracks ETHICS moral wrongness / unacceptability judgments
- this direction is related to but not reducible to negative sentiment
- small models already contain morally relevant latent variables that can be studied mechanistically

A Prompt Templates

Put all representation and judgment templates here.

B Additional Tables

Useful appendix tables:

- layerwise metrics
- causal AUROC comparison table
- bootstrap summary table

C Example Error Cases

Pull examples from `notes/error_modes.md` and `notes/latent_expressed_mismatch.md`.